Propionibacterium acnes Activates the NLRP3 Inflammasome in Human Sebocytes

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Propionibacterium acne and sebaceous glands are considered to have an important role in the development of acne. Although information regarding the activation of innate immunity by *P. acnes* in the sebaceous gland is limited, different *P. acnes* phylotypes and a higher prevalence of follicular *P. acnes* macrocolonies/biofilms in sebaceous follicles of skin biopsies from acne compared with control skin and occasionally single *P. acnes* clusters in single sebaceous glands have been detected. In this study, we investigated whether *P. acnes* activates the inflammasome in human sebaceous glands *in vivo* and *in vitro*. We found that IL-1β expression was upregulated in sebaceous glands of acne lesions. After stimulation of human sebocytes with *P. acnes*, the activation of caspase-1 and secretion of IL-1β were enhanced significantly. Moreover, knocking down the expression of NLRP3 abolished *P. acnes*-induced IL-1β production in sebocytes. The activation of the NLRP3 inflammasome by *P. acnes* was dependent on protease activity and reactive oxygen species generation. Finally, we found that NALP3-deficient mice display an impaired inflammatory response to *P. acnes*. These results suggest that human sebocytes are important immunocompetent cells that induce the NLRP3 inflammasome, and that *P. acnes*-induced IL-1β activation in sebaceous glands may have a role in combating skin infections and in acne pathogenesis.

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INTRODUCTION

Clinically relevant inflammation in acne may be initiated by disruption of the follicular epithelium followed by the spread of bacteria, including *Propionibacterium acnes*, to the dermis, leading to the development of papules, pustules, and nodulocystic lesions (Zouboulis, 2005; Kurokawa *et al.*, 2009). *P. acnes* induces the expression of proinflammatory cytokines in various cells involved in cutaneous innate immunity (Ingham *et al.*, 1992; Kim *et al.*, 2002; Graham *et al.*, 2004). Although *P. acnes* is a commensal bacterium of normal skin, it (together with the sebaceous gland) is considered to have an important role in acne development.

Sebocyte cytokine regulation has a key role in acne pathophysiology. Human sebocytes constitutively express IL-8, and can be induced to also express IL-1α, IL-1β, and IL-6 (Zouboulis *et al.*, 1998; Alestas *et al.*, 2006). The stimulation of sebocytes with *P. acnes* significantly upregulated the expression of proinflammatory cytokines (Nagy *et al.*, 2006; Clarke *et al.*, 2007; Kurokawa *et al.*, 2009). Moreover, sebocytes may act as immune competent cells capable of pathogen recognition via expression of Toll-like receptor 2, Toll-like receptor 4, CD1, and CD14 (Georgel *et al.*, 2005; Oeff *et al.*, 2006). Toll-like receptors recognize pathogen-associated molecular patterns, such as microorganisms, and then promote the transcription of proinflammatory cytokines and antimicrobial peptides (Koreck *et al.*, 2003). These findings indicate that functional induction of innate immunity in human sebocytes is important for recognizing microbial components and inducing cytokine synthesis in acne.

The innate immune system recognizes pathogens via pattern recognition receptors, such as Toll-like receptors and Nod-like receptors (NLRs). NLR members, including NLRP1, NLRP3, NLRC4 (also known as IPAF), and absent in melanoma 2 (AIM2), can mediate responses to pathogen-associated molecular patterns and subsequently form multiprotein complexes, such as the inflammasome, with the adapter apoptotic speck-like protein containing caspase recruitment domain (CARD) (Apoptosis-associated Speck-like protein containing a CARD (ASC)) and pro-caspase-1. Caspase-1 is then activated, leading to processing and secretion of inflammatory cytokines IL-1β and IL-18, resulting in inflammation *in vivo* (Martinon *et al.*,

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2002; Wilmanski *et al.*, 2008). A number of bacteria are known to activate various inflammasomes. NLRP1 and NLRP3 are mainly activated by Gram-positive bacteria, whereas Gram-negative bacteria activate the NLRC4 inflammasome (Mariathasan *et al.*, 2004; Mariathasan *et al.*, 2006; Vande Walle and Lamkanfi, 2011). The family of *P. acnes* (Gram-positive bacteria) is known to induce chronic inflammation in acne, whereas a role for inflammasome-mediated inflammation in acne pathogenesis has recently been suggested (Kistowska *et al.*, 2014; Qin *et al.*, 2014). However, knowledge on the involvement of inflammasome activation in sebocytes in *P. acnes*-induced inflammation is limited, although different *P. acnes* phylotypes and a higher prevalence of follicular *P. acnes* macrocolonies/biofilms were detected in sebaceous follicles of skin biopsies from acne compared with control skin and occasionally single *P. acnes* clusters in single sebaceous glands were also detected (Jahns *et al.*, 2012). In this study, we explored whether *P. acnes* triggers IL-1β-mediated inflammatory responses via inflammasome activation in human sebocytes.

RESULTS

IL-1β and caspase-1 expression are increased in sebaceous glands of acne lesions

We first examined IL-1β expression in skin lesions of acne patients compared with skin from healthy donors. Immunohistochemical analyses indicated that IL-1β was highly expressed in macrophages surrounding the pilosebaceous unit in acne lesions, and this was consistent with previous studies (Kistowska *et al.*, 2014). In addition, IL-1β expression was significantly upregulated in sebaceous glands in acne lesions. The percentage of positively stained sebocytes was also higher in the acne lesions than in normal skin (Figure 1a). Subsequent IL-1β processing is usually mediated by caspase-1 and, consequently, caspase-1 activation in sebaceous glands was also increased in acne lesions (Figure 1b). Therefore, we focused on sebocytes to investigate the inflammasome pathway involved in acne lesions.

P. acnes induces IL-1β processing and secretion in human sebocytes

To explore inflammasome activity in the sebocytes of acne lesions *in vitro*, we used human SZ95 sebocytes. We first examined whether SZ95 sebocytes contain the molecular components required for inflammasome assembly. Messenger RNA transcripts for the majority of components of the inflammasome were detectable in SZ95 sebocytes, similar with macrophages and keratinocytes, although NLRP1 and IPAF were only detectable in macrophages and keratinocytes (Supplementary Data online; Supplementary Figure S1 online). This finding indicated that cultured human sebocytes also contain the molecular components necessary for inflammasome assembly.

To explore whether *P. acnes* triggers IL-1β production *in vitro*, SZ95 sebocytes were exposed to living *P. acnes*. Similar with immunohistochemical analysis in acne lesions, cytoplasmic expression of IL-1β was upregulated after exposure to *P. acnes* for 24 hours (Figure 2a). Using real-time PCR, we also confirmed the elevated levels of IL-1β messenger RNA after stimulation with *P. acnes* (Figure 2b). We further analyzed the release of IL-1β in SZ95 sebocytes exposed to *P. acnes,* and found that it significantly enhanced IL-1β release in a multiplicity of infection-dependent manner (Figure 2c). Moreover, significant pro-IL-1β synthesis and subsequent mature IL-1β secretion could be detected by western blot in cell lysates and supernatants, respectively (Figure 2d).

P. acnes–induced caspase-1 activation is required for IL-1β secretion

We next investigated whether *P. acnes* treatment of human sebocytes activated caspase-1, as subsequent proteolytic activation of the IL-1β peptide is mediated by caspase-1. Caspase-1 is initially expressed as an inactive precursor that is cleaved upon stimulation, yielding active p10 and p20 subunits (Walker *et al.*, 1994; Martinon *et al.*, 2002). Compared with unstimulated sebocytes, we found that SZ95 sebocytes stimulated with *P. acnes* significantly induced the expression of active caspase-1 p20 in a multiplicity of infection-dependent manner (Figure 3a). To confirm the functional role of caspase-1 in IL-1β secretion, we generated a recombinant adenovirus expressing microRNA (mIR) specific for caspase-1 to knock down expression of this gene (Figure 3b). As shown in Figure 3c, mIR specific for caspase-1 significantly reduced IL-1β secretion by *P. acnes* stimulation, whereas a scrambled mIR did not. ELISA analysis of supernatants from *P. acnes*–stimulated SZ95 sebocytes further confirmed that *P. acnes* activation of caspase-1 has an important role in IL-1β production in SZ95 sebocytes.

P. acnes induces the release of IL-1β from human sebocytes via the NLRP3 inflammasome

Similar to the results of caspase-1, we found that knocking down the adapter protein ASC abolished the secretion and processing of IL-1β in response to *P. acnes* (Supplementary Figure S2 online). These data suggest that ASC has a role in sensing *P. acnes* for induction of IL-1β maturation. We next evaluated whether a specific inflammasome is involved in *P. acnes* induction of IL-1β. To date, four inflammasomes (NLRP3, NLRP1, NLRC4, and AIM2) have been studied extensively (Davis *et al.*, 2011). As shown in Supplementary Figure S1 online, NLRP1 and IPAF were almost undetectable in SZ95 sebocytes; therefore, we focused on NLRP3 and AIM2. To determine whether *P. acnes* activates IL-1β release via the NLRP3 or AIM2 inflammasome, we utilized a recombinant adenovirus expressing miR-NLRP3 and miR-AIM2 (Figure 4a). miR-NLRP3, but not miR-AIM2, significantly reduced the generation of active caspase-1 and consequently inhibited the secretion of mature IL-1β induced by *P. acnes* stimulation (Figure 4b). Moreover, ELISA analysis of the supernatants also confirmed that miR-NLRP3 significantly suppressed *P. acnes*–induced IL-1β secretion (Figure 4c). Taken together, these results suggest that the NLRP3 inflammasome is necessary for *P. acnes*-induced IL-1β generation and secretion in SZ95 sebocytes. To corroborate our *in vitro* data, which were obtained using SZ95 sebocytes, we have performed additional experiments with primary human

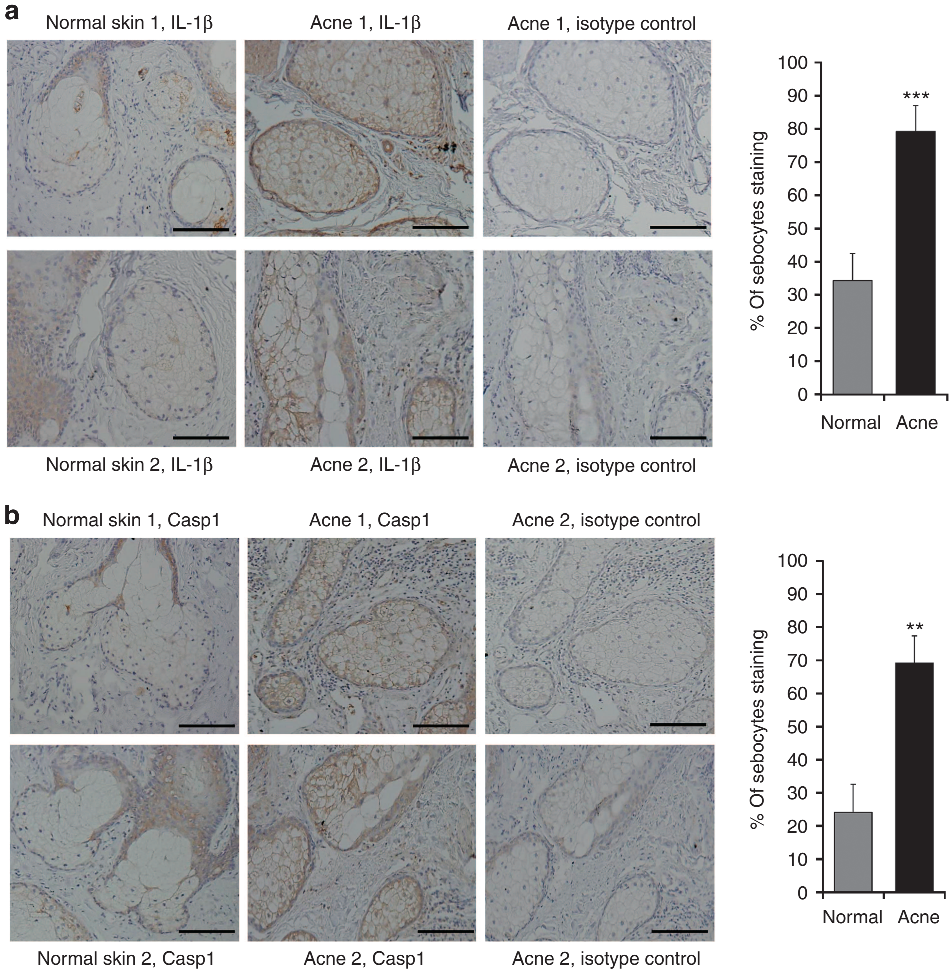

Figure 1. Expression of IL-1β and caspase-1 in sebaceous glands of acne lesions. Paraffin-embedded skin specimens of normal human facial skin ($n=4$) and facial inflammatory acne lesions ($n=6$) were immunohistochemically stained with (**a**) anti-IL-1β and (**b**) caspase-1 (Casp1). The percentage of sebocytes with cytoplasmic immunostaining was evaluated in several sebaceous glands. Bars = 100 μm. Data represent mean ± SEM. Data were analyzed by the Student's *t*-test (**$P<0.01$, ***$P<0.001$).

sebocytes from four donors and confirmed that *P. acnes* triggers IL-1β via NLRP3 and caspase-1 activation in primary human sebocytes too (Supplementary Figure S3 online).

P. acnes-induced caspase-1/IL-1β activation depends on metalloproteinase activity

We next investigated the mechanisms underlying NLRP3 inflammasome activation by *P. acnes* colonization. *P. acnes* produces extracellular enzymes, such as proteases, which may participate in acne inflammation by matrix breakdown and proteolytic detachment of follicular keratinocytes. In addition to direct proteolytic effects, proteases derived from *P. acnes* could be potential activators of innate immune responses and inflammation (HoeZer, 1977; Ingram *et al.*, 1983; Lee *et al.*, 2010). To investigate whether *P. acnes*–derived proteases are involved in inflammasome activation, *P. acnes* was incubated at 65 °C for 30 minutes to inactivate proteases (Ahern and Klibanov, 1987; Kauffman *et al.*, 2006). Using western blots, we found that heat treatment downregulated the generation of active caspase-1 and IL-1β by viable *P. acnes* (Figure 5a). In addition, heat-treated *P. acnes* almost completely blocked IL-1β release according to ELISA analysis (Figure 5b). We further tested the effects of a specific protease inhibitor, EDTA. Pre-incubation of *P. acnes* with EDTA impaired the activation of caspase-1 and inhibited the release of IL-1β in a concentration-dependent manner (Figure 5c and d). In contrast, the specific cysteine inhibitor E64 did not block the release of IL-1β (data not shown). These data indicate that activation of the NLRP3 inflammasome by *P. acnes* was dependent on *P. acnes*–derived metalloprotease activity.

ROS generation is required for *P. acnes*-induced IL-1β secretion from sebocytes

We analyzed the mechanism leading to inflammasome activation upon *P. acnes* exposure. Many bacteria activate the NLRP3 inflammasome through potassium (K^+) efflux and reactive oxygen species (ROS) generation (Tschopp and Schröder, 2010; Koizumi *et al.*, 2012). To explore the role

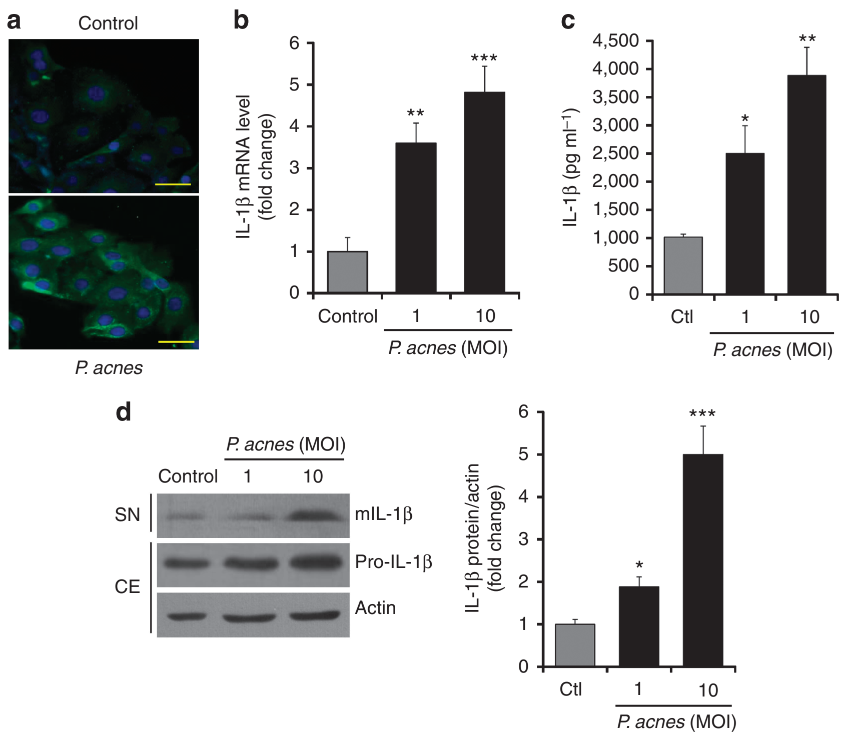

Figure 2. *P. acnes*-induced IL-1β processing and secretion in SZ95 sebocytes. (**a**) Immunofluorescence labeling of IL-1β (green) in human SZ95 sebocytes after exposure to *Propionibacterium acnes* at a multiplicity of infection (MOI) of 10 for 24 hours. Nuclei were counterstained with 4,6-diamidino-2-phenylindole (blue). Bars = 100 μm. (**b**) Quantitative reverse-transcriptase–PCR analysis of IL-1β messenger RNA expression after *P. acnes* stimulation (MOI of 1 and 10). (**c**) Mature IL-1β secretion in supernatants at a MOI after *P. acnes* stimulation was assessed using ELISA. (**d**) Pro-IL-1β protein in the cell extracts (CE) and mature IL-1β protein (mIL-1β) in supernatants (SN) were determined by western blot. Data represent mean ± SEM. Data were analyzed by the Student's *t*-test ($n = 3$; $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$).

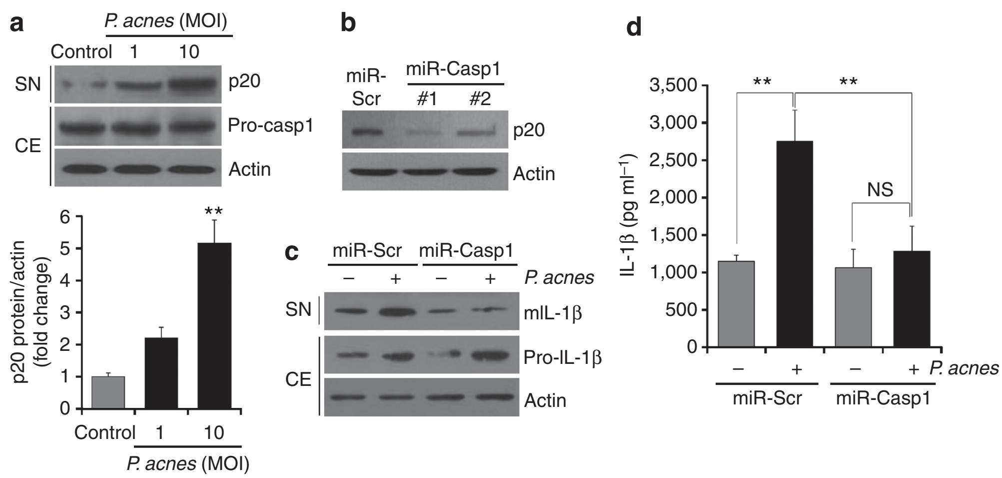

Figure 3. *P. acnes*-induced IL-1β secretion requires caspase-1 activation in SZ95 sebocytes. (**a**) The expression of the active caspase-1 p20 subunit (p20) was determined in SZ95 sebocytes by western blot after exposure to *Propionibacterium acnes*. The p20 protein levels were expressed relative to the actin control (lower panel). (**b**) SZ95 sebocytes were transduced with an adenovirus expressing caspase-1 specific microRNA (miR-Casp1) and scrambled microRNA (miR-Scr) as the control. (**c**) Supernatants (SN) and cell extracts (CE) from miR-Scr- and miR-Casp1-transduced sebocytes after *P. acnes* (multiplicity of infection (MOI) of 10) stimulation were analyzed for expression of IL-1β by western blot. (**d**) Supernatants were also analyzed for IL-1β by ELISA. NS, no significant difference. Data represent mean ± SEM. Data were analyzed by the Student's *t*-test ($n = 3$; $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$).

of K^+ efflux and ROS in *P. acnes*-induced IL-1β maturation in SZ95 sebocytes, specific inhibitors were used to block their activity. There was no significant change in IL-1β secretion in the presence of glibenclamide, a selective inhibitor of ATP-sensitive K^+ channels (Figure 5a). However, the presence of N-acetyl-L-cysteine, an inhibitor of ROS generation,

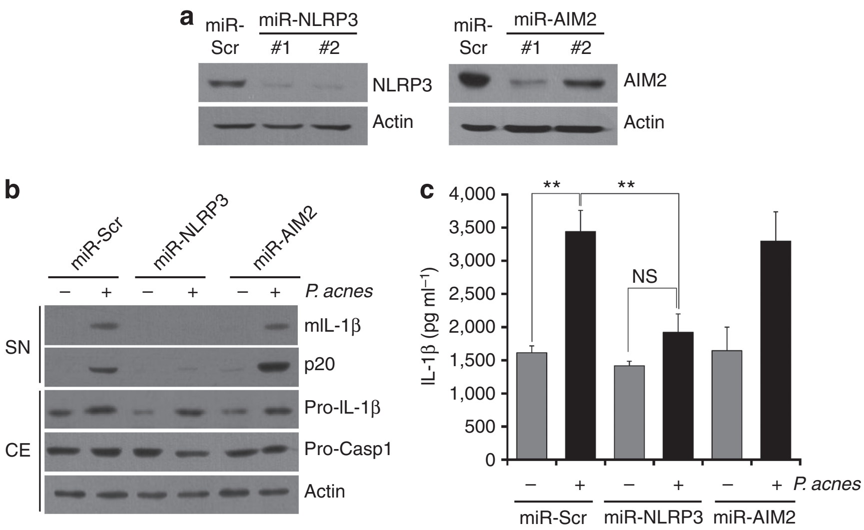

Figure 4. *P. acnes* induction of IL-1β is NLRP3-dependent in SZ95 sebocytes. (**a**) SZ95 sebocytes were transduced with an adenovirus expressing microRNA specific for NLRP3 (miR-NLRP3) and absent in melanoma 2 (AIM2) (miR-AIM2). (**b**) MicroRNA-transduced SZ95 sebocytes were stimulated with *Propionibacterium acnes* (multiplicity of infection (MOI) of 10) for 24 hours, and then supernatants (SN) and cell extracts (CE) were analyzed by western blot. (**c**) Supernatants were also analyzed for IL-1β by ELISA. NS, no significant difference. Data represent mean ± SEM. Data were analyzed by the Student's *t*-test ($n=3$; $^{**}P<0.01$).

abrogated production of *P. acnes*-induced IL-1β in a dose-dependent manner (Figure 5b). Thus, ROS generation is important for *P. acnes*-induced NLRP3 inflammasome activation in sebocytes.

P. acnes-induced inflammation is reduced in NALP3-deficient mice

To confirm the role of the inflammasome in *P. acnes*-induced inflammation *in vivo*, we used NLRP3-deficient mice and their wild-type littermates. To induce a model of *P. acnes* inflammation, we injected living *P. acnes* intradermally into mouse ears. In wild-type mice, significant cutaneous erythema and ear swelling were observed in *P. acnes*–injected ears 24 hours after challenge, but neither symptom was induced by phosphate-buffered saline (PBS) injection (Figure 6a). Moreover, we found that NLRP3-deficient mice showed a significant decrease in ear thickness compared with wild-type mice (Figure 6b). Histological observation also showed a significant decrease in the number of infiltrating inflammatory cells after *P. acnes* injection in NLRP3-deficient mice (Figure 6c). To determine the specific role of sebaceous glands related with these findings, we performed immunohistochemical detection of caspase-1 and IL-1β in the sebaceous glands of the specimens. As shown in Figure 6d, inoculation of *P. acnes* increased caspase-1 and IL-1β expression in sebaceous glands of the mouse skin compared with PBS-injected skin, while a marked decrease in these expressions was obvious in the skin of NLRP3-deficient mice. In addition, we demonstrated that NLRP3 and caspase-1 were expressed and colocalized in the sebaceous glands of mouse skin after injection of *P. acnes* (Supplementary Figure S4 online). These observations are consistent with our *in vitro* data and indicate that the NLRP3 inflammasome triggers an inflammatory reaction in the sebaceous glands at the site of *P. acnes* colonization.

DISCUSSION

Sebocytes (together with keratinocytes) are major components of the pilosebaceous unit and may act as immune competent cells activated by *P. acnes* (Nagy *et al.*, 2006; Jahns *et al.*, 2012). With the current work, we provided evidence that human sebocytes contain the necessary elements to form an inflammasome and that *P. acnes* induces IL-1β release via NLRP3 inflammasome activation. These findings also suggest that acne is an autoinflammatory disease characterized by aberrant regulation of the innate immune system. Previous studies have shown that acne may be capable of triggering innate immune responses via the inflammasome. Several autoinflammatory diseases, such as Pyogenic Arthritis, Pyoderma gangrenosum, and Acne (PAPA) syndrome, result from inflammasome formation and are characterized by severe acne (Shoham *et al.*, 2003; Chen *et al.*, 2011; Nguyen *et al.*, 2013). NLRP3 inflammasome activation was recently reported to mediate IL-1β regulation in human monocytes in response to *P. acnes* (Kistowska *et al.*, 2014; Qin *et al.*, 2014).

Owing to the important role of IL-1 in the induction of inflammatory responses, aberrant activity of cytokines and inflammasomes is involved in the pathogenesis of many skin diseases, such as psoriasis, vitiligo, and contact dermatitis (Jin *et al.*, 2007; Johansen *et al.*, 2007; Watanabe *et al.*, 2007). IL-1 is known to have a key role in the early development of acne lesions (Zouboulis, 2001; Koreck *et al.*, 2003). IL-1 is a potent inducer of cell infiltration and proinflammatory cytokine release and it additionally promotes comedone formation after hyper-proliferation of follicular keratinocytes (Ingham *et al.*, 1992; Guy and Kealey, 1998; Beylot *et al.*, 2014). It is known that keratinocytes and macrophages in acne lesions are a major source of IL-1 induced by *P. acnes* (Kim *et al.*, 2002; Qin *et al.*, 2014). However, the significance of sebocytes in *P. acnes*-induced IL-1 activation still remained

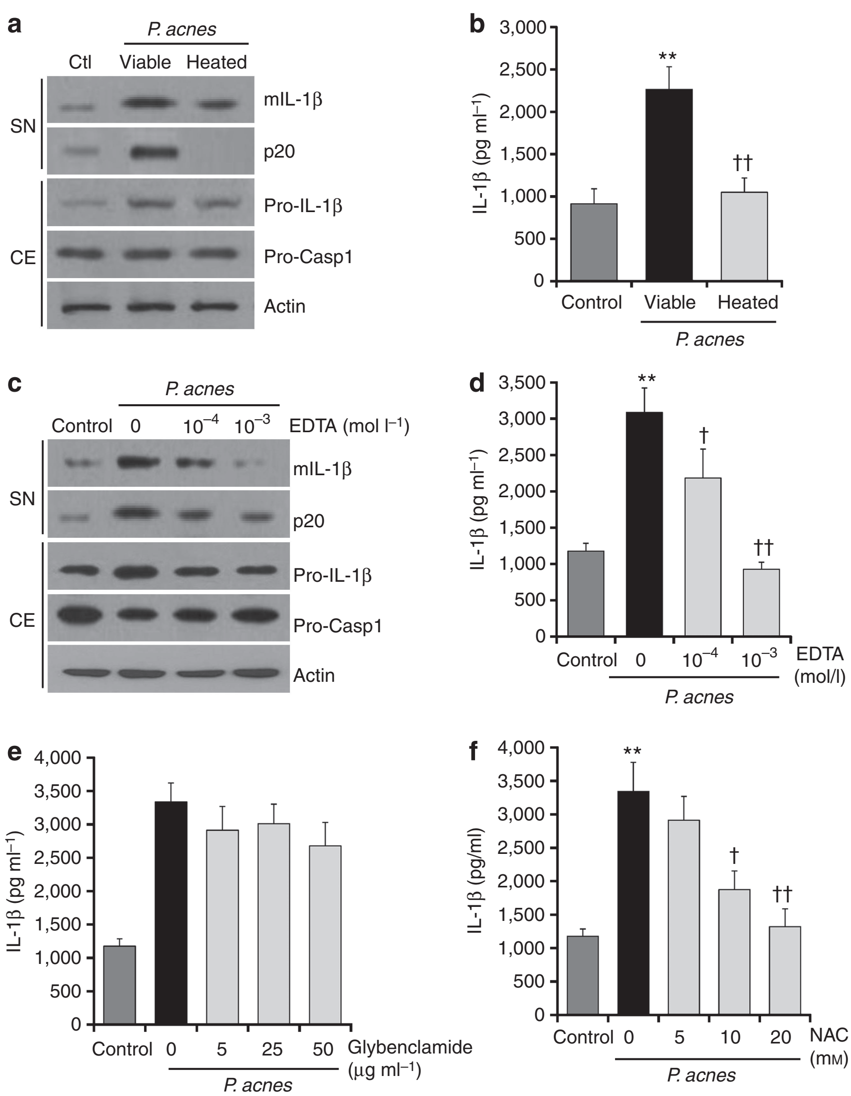

Figure 5. Metalloprotease activity and reactive oxygen species generation are required for *P. acnes*-induced inflammasome activation. *Propionibacterium acnes* (multiplicity of infection (MOI) of 10) was incubated at 65 °C for 30 minutes and then applied to SZ95 sebocytes. (**a**) Supernatants (SN) and cell extracts (CE) were analyzed by western blotting. (**b**) Supernatants were also analyzed for IL-1β by ELISA. (**c**, **d**) SZ95 sebocytes were treated with different concentrations of EDTA and were then stimulated with *P. acnes*. After SZ95 sebocytes were exposed to *P. acnes* in the presence of (**e**) glibenclamide and (**f**) N-acetyl-L-cysteine (NAC), IL-1β release in supernatants was measured by ELISA. Data represent mean ± SEM. Data were analyzed by the Student's *t*-test ($n=3$; $^{**}P<0.01$ vs *P. acnes*-unstimulated cells; $^{\dagger}P<0.05$, $^{\dagger\dagger}P<0.01$ vs only *P. acnes*-treated cells).

unclear. A previous study treated cultured sebocytes with *P. acnes* and found no effect on IL-1α, but it did observe upregulated expression of TNF-α and CXCL8 (Nagy *et al.*, 2006). However, IL-1α was shown to be produced and released by human sebocytes under stress conditions (Zouboulis *et al.*, 1998). As shown in Figures 1 and 2, we found that *P. acnes* stimulates the generation and secretion of IL-1β by sebaceous glands in acne lesions *in vivo* and cultured sebocytes *in vitro*. On the other hand, different *P. acnes* phylotypes have been detected in sebaceous follicles of acne patients (Jahns *et al.*, 2012) and these vary in their capacity to stimulate an inflammatory response in human sebocytes (Nagy *et al.*, 2006). These data can explain the varying findings in our work and that of Nagy *et al.* (2006).

The exact mechanism involved in *P. acnes*-induced IL-1β secretion in sebocytes remains unclear. However, we demonstrated that proteases are necessary for NLRP3 activation of caspase-1 and subsequent IL-1β release. *P. acnes* is known to produce exogenous proteases and elicit cellular responses to mediate inflammation via proteinase activated receptor 2 in acne lesions (Lee *et al.*, 2010; Beylot *et al.*, 2014). Functional proteinase activated receptor 2 is expressed by sebocytes as well as keratinocytes and inflammatory cells (Rattenholl and Steinho, 2003). We demonstrated that *P. acnes*–derived proteinase activity is required for *P. acnes*–mediated inflammasome activation, and that heat or EDTA prevented NLRP3 inflammasome-dependent inflammation. Therefore, the pharmacological regulation of protease expression

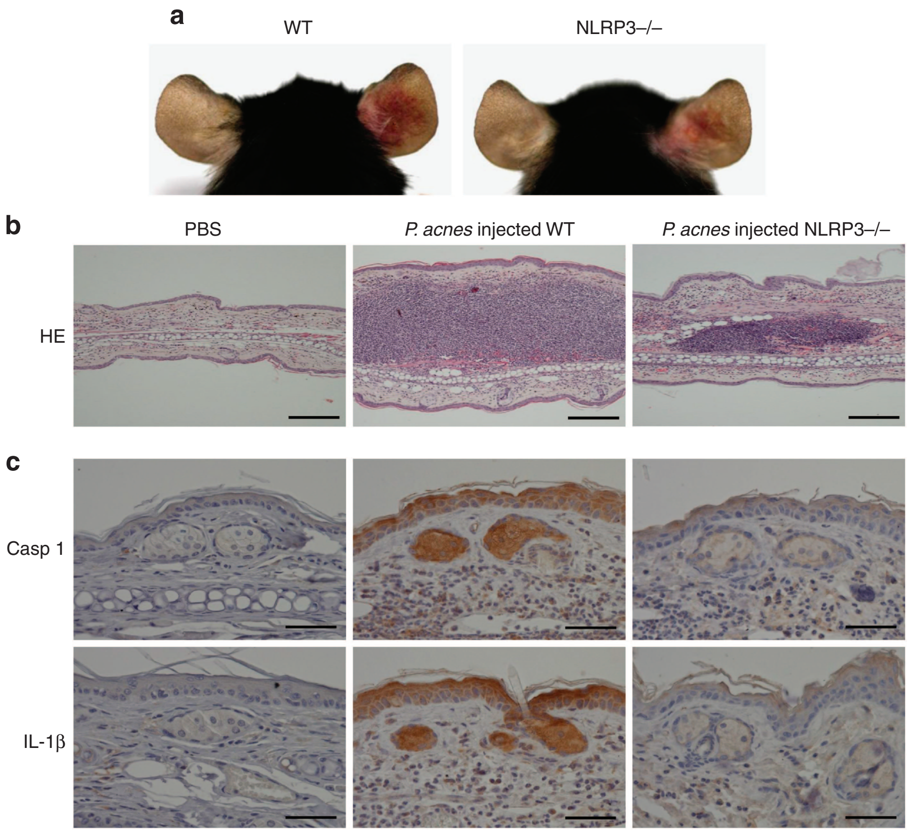

Figure 6. *P. acnes*-induced inflammation is reduced in NALP3-deficient mice. (**a**) Wild-type (WT) ($n=10$) and NLRP3-deficient mice ($n=10$) were injected intradermally with 10^7 colony-forming units of *P. acnes* (right ear) or an equal volume of phosphate-buffered saline (PBS) (left ear). Inflammation-induced redness of the ear was visualized 24 hours after injection. (**b**) Hematoxylin and eosin (HE)-stained frozen tissue sections of the ear injected with PBS alone or *P. acnes* in wild-type (WT) mice or with *P. acnes* in NLRP3-deficient mice. Bars = 200 μm. (**c**) Each specimen was immunohistochemically stained with anti-caspase-1 (Casp1) and IL-1β monoclonal antibodies. Bars = 50 μm.

may provide a target for the treatment of inflammatory acne lesions.

On the other hand, K^+ efflux and ROS are common triggers of NLRP3 inflammasome activation. Extracellular ATP released from infected cells can lead to K^+ efflux, release of cathepsins, and subsequent induction of NLRP3 inflammasome activation (Di Virgilio, 2007; Hornung *et al.*, 2009). However, in this study, sebocytes incubated with K^+ channel inhibitors did not show *P. acnes*-induced IL-1β activation, suggesting that *P. acnes* activates the NLRP3 inflammasome independently of K^+ efflux in sebocytes. Other important crucial elements for NLRP3 activation include ROS. ROS production results in NLRP3 inflammasome activation through the release of thioredoxin-interacting protein (Dostert *et al.*, 2008; Tschopp and Schröder, 2010; Zhou *et al.*, 2010). Our study demonstrated that IL-1β secretion from SZ95 sebocytes exposed to *P. acnes* was reduced by ROS inhibitors. Proteases and ROS are commonly observed at inflammation sites, and it has been previously shown that proteases stimulate ROS release from various cells (Speer *et al.*, 1984; Weiss and Regiani, 1984; Ossanna *et al.*, 1986; Aoshiba *et al.*, 2001). Therefore, we hypothesize that *P. acnes*–secreted proteases result in ROS generation, after which IL-1β is released via NLRP3 inflammasome activation in sebocytes.

This study provides first evidence that human sebocytes constitutively express inflammasome proteins, and *P. acnes* triggers NLRP3 inflammasome activation, resulting in IL-1β release from sebocytes. These findings indicate that inflammasome-dependent activation and secretion of IL-1β is not restricted to macrophages in acne. This new finding for *P. acnes* suggests a functional role of sebaceous glands in innate immunity (Georgel *et al.*, 2005) that may be important in combating skin infections and during acne pathogenesis (Alestas *et al.*, 2006). Understanding the mechanisms of inflammasome activation in sebaceous glands could lead to new approaches in the prevention and treatment of acne.

MATERIALS AND METHODS

Reagents

P. acnes (KCTC 3314) was obtained from the Korean Collection for Type Cultures (KCTC, Daejeon, Korea). EDTA was obtained from Biosesang (Seongnam, Korea) and glibenclamide and N-acetyl-L-cysteine were purchased from InvivoGen (San Diego, CA) and Sigma-Aldrich (St Louis, MO), respectively. For western blotting and immunohistochemical analysis, we used the following specific antibodies: IL-1β, AIM2 (Abcam, Cambridge, MA), caspase-1 (Cell Signaling Technology, Danvers, MA), NLRP3, ASC (Adipogen, San Diego, CA), and actin (Santa Cruz Biotechnologies, Santa Cruz, CA).

Immunohistochemistry

Skin samples from six patients (four men and two women) with acne vulgaris aged 20–28 years were used for this study. Punch biopsy specimens (3-mm thick) were obtained from the inflammatory acne lesions of the face. Normal human facial skin specimens were obtained surgically from the face of four patients (two male and two female), aged 26–48 years, undergoing plastic surgery. Tissue samples were fixed with 10% formaldehyde, embedded in paraffin, and cut into 4-μm-thick sections. The sections were deparaffinized in xylene and then rehydrated using an alcohol series. The primary antibody was diluted 1:200. The sections were then incubated with secondary antibody at room temperature for 30 minutes. The sections were incubated with diaminobenzidine tetrachloride solution at room temperature for 5 minutes and counterstained with Mayer's hematoxylin. Between 40 and 50 sequential sections, representing every fifth section cut, were examined from each skin specimen. Finally, immunostained and unstained cytoplasms in the sebaceous glands were counted. All slides were stained at the same time. All human skin samples were obtained with the written informed consent of the donors and in accordance with the ethical committee approval process of the Institutional Review Board of Chungnam National University Hospital.

Cell culture

Immortalized human SZ95 sebocytes (Zouboulis *et al.*, 1999) showing morphologic, phenotypic, and functional characteristics of normal human sebocytes were cultured in Sebomed medium (Biochrom, Berlin, Germany) supplemented with 10% fetal bovine serum (Gibco BRL, Rockville, MD) and 5 ng ml^{-1} recombinant human epidermal growth factor (Invitrogen, Grand Island, NY). Cultures were maintained at 37 °C in a humidified atmosphere containing 5% CO_2.

Bacterial culture

P. acnes were grown in the brain–heart infusion agar (Oxoid, Basingstoke, UK) to mid-log phase at 37 °C under anaerobic conditions using the Bactron Anaerobic Chamber system (SHELLAB, Cornelius, OR). Gas conditions were 5% H_2, 5% CO_2, and 90% N_2. Bacterial cultures were diluted at a final concentration of 1 and 10 multiplicity of infection and used to stimulate cells.

Immunofluorescence

SZ95 sebocytes were grown on coverslips, fixed via incubation with 4% paraformaldehyde for 20 minutes, and permeabilized via incubation with 0.1% Triton X-100 in PBS for 10 minutes at room temperature. Cells were then incubated overnight at 4 °C with anti-IL-1β antibody and then for 1 hour at room temperature with FITC-conjugated secondary antibody, and they were finally visualized under a fluorescence microscope (Olympus Corporation, Tokyo, Japan).

Real-time PCR

Aliquots of a reverse transcription mixture were analyzed by quantitative real-time reverse-transcriptase–PCR using specific primer sets. The following primer sequences were used: IL-1β, forward 5′- ACG AATCTCCGACCACCACTA-3′ and reverse 5′-TCCATGGCCACAACA ACTGA-3′. For quantitative real-time reverse-transcriptase–PCR, SYBR Green real-time PCR master mix (Invitrogen, Carlsbad, CA) was used according to the manufacture's protocol. All reactions were independently repeated at least three times to ensure the reproducibility of the results.

ELISA

To measure cytokine release, supernatants were collected and frozen until used for ELISA assays. IL-1β was detected using ELISA kits from MyBioSource (San Diego, CA) (MBS888258) (sensitivity <2 pg and <0.2 pg per ml, respectively). ELISAs were performed according to the manufacturer's instructions.

Western blot analysis

Cells were lysed in Proprep solution (Intron, Deajeon, Korea). Samples were run on SDS-polyacrylamide gels, transferred onto nitrocellulose membranes, and incubated with appropriate primary antibodies. Blots were then incubated with peroxidase-conjugated secondary antibodies and visualized by enhanced chemiluminescence (Intron).

Knock down of gene expression

miR specific for caspase-1, NLRP3, and AIM2 target sequences were designed using Invitrogen's RNAi Designer (Invitrogen, Carlsbad, CA). The double-stranded DNA oligonucleotides were synthesized and cloned into the parental vector pcDNA6.2-GW/EmGFP-miR (Invitrogen, Carlsbad, CA). The expression cassette for miR was moved into the pENT/CMV vector and adenovirus was made as previously reported (Shi *et al.*, 2013). The miR sequence is provided in Supplementary Table S1 online.

Animal studies

We used 7-week-old NLRP3-deficient mice (B6N.129-Nlrp3^{tm2Hhf}/J, The Jackson Laboratory, Bar Harbor, ME) (Brydges *et al.*, 2009) and C57BL/6 mice (Orient Bio, Gyeonggi, Korea) in these experiments. The experiment was approved by Chungnam National University's Institutional Animal Care and Use Committees and was conducted in accordance with the guidelines for the Care and Use of Laboratory Animals. Aliquots (20 ml) of living *P. acnes* (1×10^7 colony-forming units) suspended in PBS were intradermally injected in the central portion of the right ear. As a control, an equal volume of PBS was injected into the left ear of the same mice. For histological observation, the ear was fixed with 10% formaldehyde, embedded in paraffin, sectioned, and stained with hematoxylin and eosin.

Statistical analysis

All experiments were repeated at least three times with separate batches of cells. One-way analysis of variance was used to compare variances within and among groups. Data were evaluated statistically using *post hoc* two-tailed Student's *t*-tests. Statistical significance was set at $P<0.05$.

CONFLICT OF INTEREST

The authors state no conflict of interest.

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SUPPLEMENTARY MATERIAL

Supplementary material is linked to the online version of the paper at https://www.sciencedirect.com/journal/journal-of-investigative-dermatology

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